



AI

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In this section we gain insight into the real sentiments behind the surge in AI popularity. Is it making a real impact in the way developers work or is it all hype?

[3.1. Sentiment and usage →](#)

[3.2. Developer tools →](#)

[3.3. Efficacy and Ethics →](#)

Sentiment and usage 3.1

AI tools in the development process

76% of all respondents are using or are planning to use AI tools in their development process this year, an increase from last year (70%). Many more developers are currently using AI tools this year, too (62% vs. 44%).

Do you currently use AI tools in your development process? *

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All Respondents Professional Developers Learning to Code Other Coders



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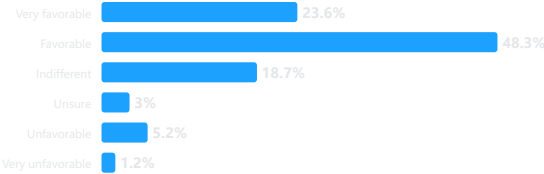
Responses: 60,907 (93.1%)

AI tool sentiment

72% of all respondents are favorable or very favorable of AI tools for development. This is lower than last year's favorability of 77%; a decline in favorability could be due to disappointing results from usage.

How favorable is your stance on using AI tools as part of your development workflow?

All Respondents Professional Developers Learning to Code Other Coders



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Responses: 45,873 (70.1%)

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Benefits of AI tools

81% agree increasing productivity is the biggest benefit that developers identify for AI tools. Speeding up learning is seen as a bigger benefit to developers learning to code (71%) compared to professional developers (61%).

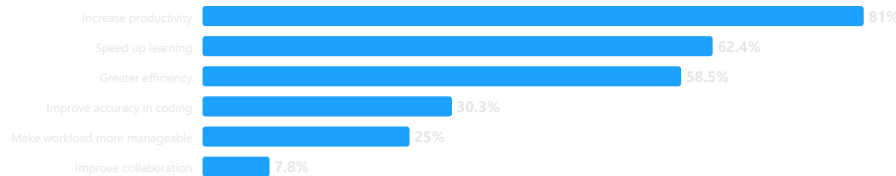
? For the AI tools you use as part of your development workflow, what are the MOST important benefits you are hoping to achieve? Please check all that apply.

All Respondents

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[Download](#) [Share](#)Responses: **36,894 (56.4%)**

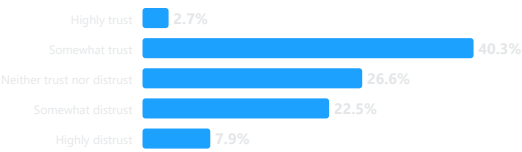
Accuracy of AI tools

Similar to last year, developers remain split on whether they trust AI output: 43% feel good about AI accuracy and 31% are skeptical. Developers learning to code are trusting AI accuracy more than their professional counterparts (49% vs. 42%).

How much do you trust the accuracy of the output from AI tools as part of your

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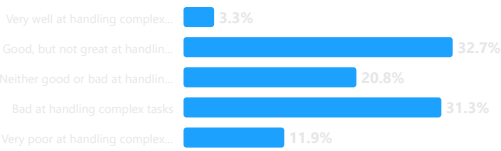
Responses: 37,302 (57%)

AI tools' ability to handle complex tasks

The development workflow is complicated, as the abundance of questions and answers on Stack Overflow can attest to. Almost half (45%) of professional developers believe AI tools are bad or very bad at handling complex tasks.

How well do the AI tools you use in your development workflow handle complex tasks?

All Respondents Professional Developers Learning to Code Other Coders

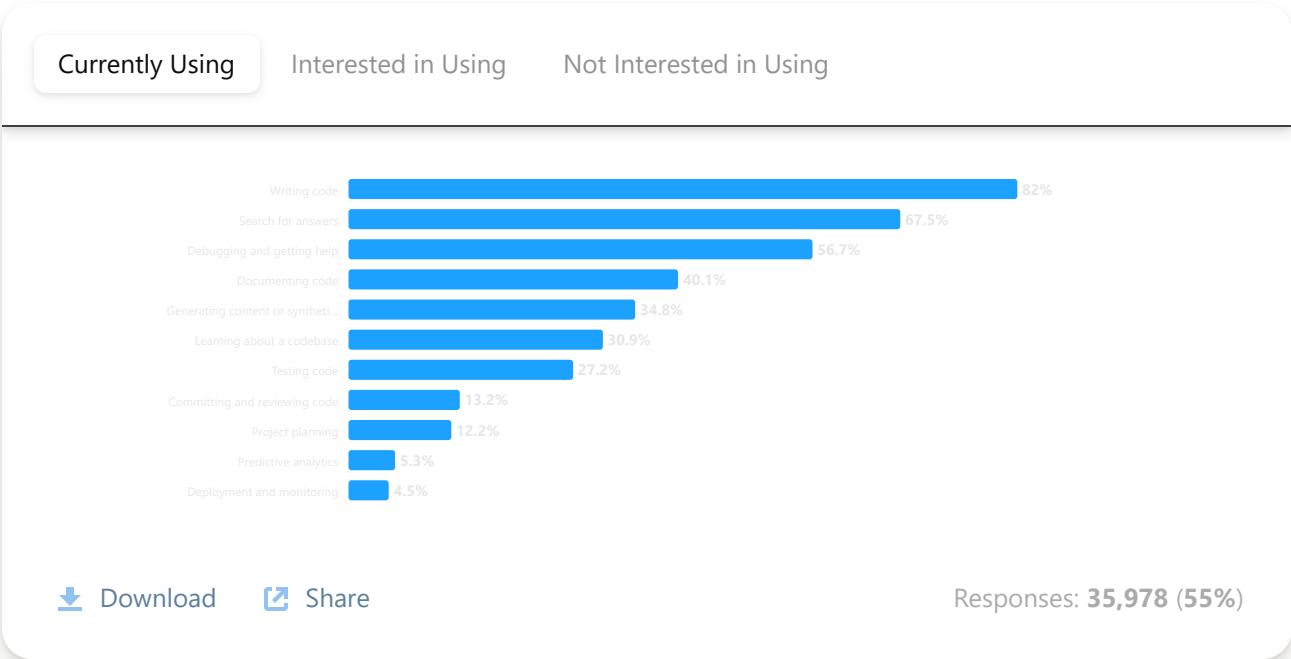


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Responses: 37,021 (56.6%)

Developers currently using AI tools mostly use them to write code (82%) and those who are interested but not yet using AI tools are mostly curious about testing code (46%). Developers with experience can trust AI tools to help write code to get started but perhaps know testing is a complex step best left to traditional processes.

? Which parts of your development workflow are you currently using AI tools for and which are you interested in using AI tools for over the next year? Please select all that apply.



AI tools next year

In the next year, most developers agree that AI tools will be more integrated mostly in the ways they are documenting code (81%), testing code (80%), and writing code (76%).

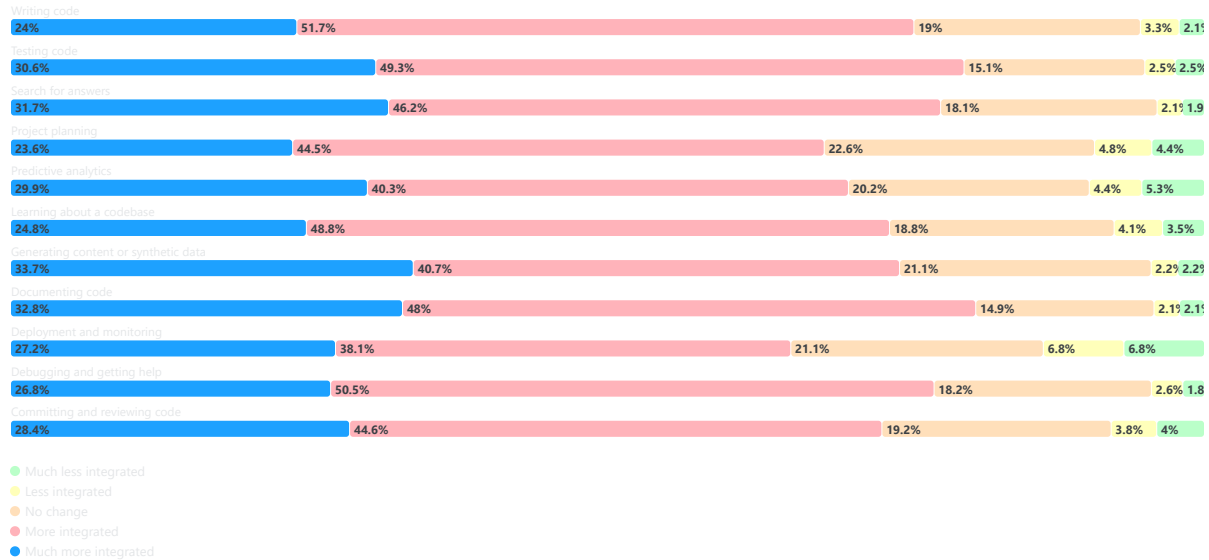
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Responses: 34,168 (52.2%)

Efficacy and Ethics 3.3

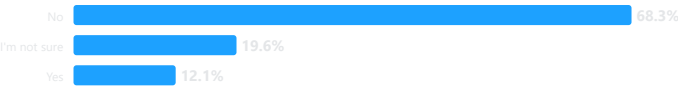
Are AI tools a threat to your job

70% of professional developers do not perceive AI as a threat to their job.

Do you believe AI is a threat to your current job?

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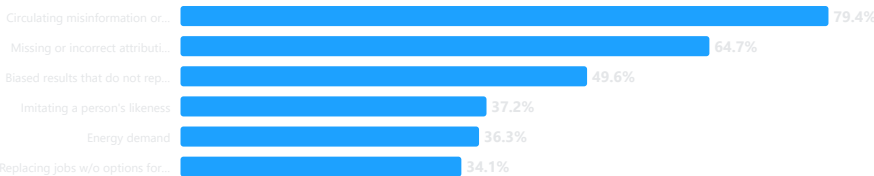
Responses: 36,328 (55.5%)

Most important ethical issues for AI

Misinformation and disinformation in AI results are a top ethical concern in this year's survey for 79% of developers. Source attribution is the next top ethical concern (65%) which [Stack Overflow is working to protect for our community members](#).

Which AI ethical responsibilities are most important to you? Select all that apply.

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Responses: 33,701 (51.5%)

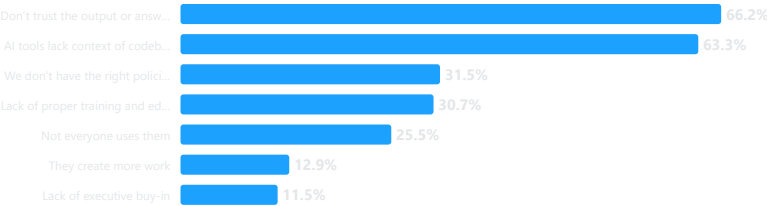
Challenges with AI at work

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Professional developers agree the issue is not user error: twice as many professionals cite lack of trust or understanding the codebase as the top challenge of AI tools compared to proper training.

? What are the challenges to your company/whole team using AI code assistants or GenAI tools? Select all that apply.

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Responses: **30,661** (46.9%)

AI at work in one year

Developers most frequently hypothesize the amount of time they spend on tasks with AI tools and the complexity of those tasks will be different one year from now.

By grouping open-ended responses into topics based on phrase patterns, we can see themes amongst developer responses.

? Please describe how you would expect your workflow to be different, if at all, in 1 year as a result of AI advancements.

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Responses: 15,136 (23.1%)

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